Quiz03 Practice Questions

Stat 230 - Spring 2026

Practice questions for each skill on the Quiz03 Review sheet. The data sets are ones we've seen in class or on the homework, but the models and questions are new.

1. Express a regression model using proper notation

Example question The fake news data includes `negative`, the percent of words in the article with negative sentiment, and `title_has_excl`, whether the title has an exclamation point. The model below allows the relationship between `negative` and the chance that an article is fake to be different for titles with and without an exclamation point.

```
news_int <- glm(type_num ~ negative * title_has_excl, data = news, family = "binomial")
tidy(news_int)
```

```
# A tibble: 4 x 5
  term                        estimate std.error statistic  p.value
  <chr>                          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)                   -1.95      0.523    -3.72  0.000199
2 negative                       0.404     0.153     2.64  0.00841
3 title_has_exclTRUE             3.57      2.27      1.57  0.116
4 negative:title_has_exclTRUE   -0.289     0.562    -0.515 0.607
```

(a) Using proper statistical notation, write down the theoretical model that was fit, on both the log-odds scale and the probability scale. Define all variables.

(b) Write down the fitted log-odds equation for titles **with** an exclamation point and for titles **without** one.

Example question In the budworm experiment, batches of 20 moths of each sex were exposed to different doses of a toxin, and the number of moths in each batch that died was recorded. The model below uses `sex` and $\log_2$(dose), and their interaction.

```r
bud_int <- glm(cbind(numdead, N - numdead) ~ sex * log2(dose),
               data = budworm, family = "binomial")
tidy(bud_int)
```

```
# A tibble: 4 x 5
  term            estimate std.error statistic      p.value
  <chr>              <dbl>     <dbl>     <dbl>        <dbl>
1 (Intercept)       -2.99      0.553    -5.42  0.0000000609
2 sexM               0.175     0.778     0.225 0.822
3 log2(dose)         0.906     0.167     5.42  0.0000000589
4 sexM:log2(dose)    0.353     0.270     1.31  0.191
```

(a) What is Y_i, and what is n_i? Using proper notation, write down the theoretical binomial logistic regression model that was fit. Define all variables.

(b) Write down the fitted log-odds equation for female moths and for male moths.

(c) Why is $\log_2(\text{dose})$ a convenient scale for this predictor? What does a 1-unit increase in $\log_2(\text{dose})$ mean in terms of the dose?

2. Interpret the coefficients of a model in context

Example question Below is a model for whether an applicant was accepted to medical school, using their undergraduate `GPA` and `MCAT` score.

```r
med_glm <- glm(Acceptance ~ GPA + MCAT, data = medschool, family = "binomial")
tidy(med_glm)
```

```
# A tibble: 3 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  -22.4       6.45      -3.47 0.000527
2 GPA            4.68      1.64       2.85 0.00439
3 MCAT           0.165     0.103      1.59 0.111
```

(a) Interpret the coefficient for `GPA` on the odds scale, for a **0.1**-point increase in GPA.

(b) Interpret the coefficient for `MCAT` on the odds scale, for a **5**-point increase in MCAT score.

(c) Does the intercept have a meaningful interpretation here? Explain.

Example question Below is the binomial logistic regression model for the number of successes (out of 4 trials) on the communication task from the language exposure study. `Bilingual` is the reference level for `language`.

```
lang_glm <- glm(success/trials ~ sex + kbit + ppvt + maternal_ed + age + language,
                data = language, family = "binomial", weights = trials)
tidy(lang_glm)
```

```
# A tibble: 8 x 5
  term                estimate std.error statistic p.value
  <chr>                  <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)           4.82       1.76      2.75   0.00600
2 sex                   0.185      0.282     0.655  0.513
3 kbit                  0.00953    0.0102    0.932  0.352
4 ppvt                 -0.0185     0.0108   -1.71   0.0875
5 maternal_ed           0.142      0.351     0.404  0.686
6 age                  -0.515      0.165    -3.12   0.00181
7 languageExposed      -0.0167     0.377    -0.0443 0.965
8 languageMonolingual  -1.07       0.353    -3.03   0.00247
```

(a) Interpret the coefficient for `age` on the odds scale. (Age is measured in years.)

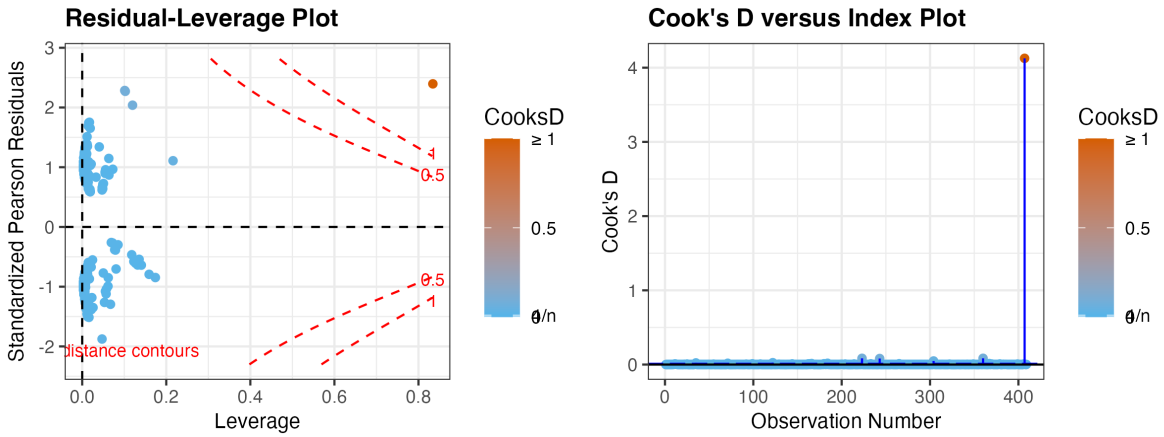
(b) Interpret the coefficient for `languageExposed` on the odds scale.

(c) We saw in class that these data are overdispersed. Would your interpretations in (a) and (b) change if we used the quasibinomial model instead? Explain.

3. Check necessary conditions for a regression model and discuss the implications of a model deficiency

Example question Below are the leverage and Cook's distance plots for the smartphone and sleep model, `good_sleep ~ phone_dependence + age + bmi + smoking + drinking + phone_before_bed` ($n = 409$), and the three students with the largest Cook's distances.

```
sleep_glm <- glm(good_sleep ~ phone_dependence + age + bmi + smoking + drinking + phone_befo
                 data = sleep, family = "binomial")
resid_panel(sleep_glm, plots = c("lev", "cookd"), type = "pearson")
```




```
sleep_aug <- augment(sleep_glm, data = sleep)
sleep_aug |>
  arrange(desc(.cooksd)) |>
  select(good_sleep, phone_dependence, age, bmi, .hat, .cooksd) |>
  head(3)
```

```
# A tibble: 3 x 6
  good_sleep phone_dependence   age   bmi  .hat .cooksd
       <int>            <int> <int> <dbl> <dbl>   <dbl>
1          1               15    76  1.04 0.834  4.13
2          1               16    16 19.4  0.102  0.0841
3          1               16    18 20.2  0.101  0.0839
```

(a) Which student has high leverage? Use the table to explain why.

(b) Is this student influential? Explain.

(c) The students in this study were all ages 15 to 21. What would you do with this observation?

(d) Why can't we use the Pearson χ^2 goodness-of-fit test to check this model?

Example question Below is a binomial logistic regression model for the moths data, using `Distance` and `Morph` **without** an interaction, along with the Pearson χ^2 statistic and an empirical logit plot.

```
moth_add <- glm(Removed/Placed ~ Distance + Morph, weights = Placed,
                data = moths, family = "binomial")
tidy(moth_add)
```

```
# A tibble: 3 x 5
  term        estimate std.error statistic    p.value
  <chr>          <dbl>     <dbl>     <dbl>      <dbl>
1 (Intercept) -0.733     0.151       -4.85 0.00000127
2 Distance     0.00531   0.00400      1.33 0.184
3 Morphlight  -0.404     0.139       -2.90 0.00374
```

```
X2 <- sum(residuals(moth_add, type = "pearson")^2)
X2
```

```
[1] 24.79796
```

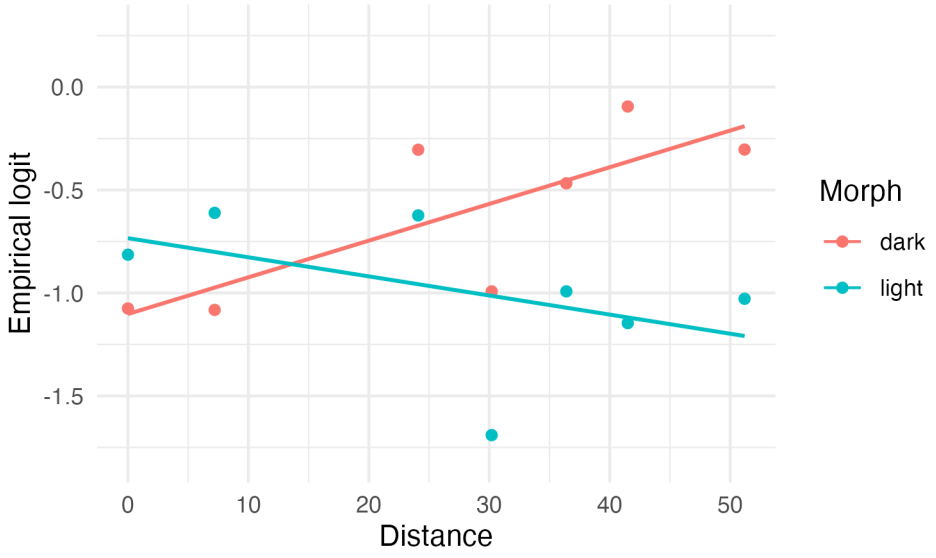
```
df.residual(moth_add)
```

```
[1] 11
```

```
1 - pchisq(X2, df = df.residual(moth_add))
```

```
[1] 0.009757989
```

```
moths |>
  mutate(emp_logit = log((Removed + 0.5) / (Placed - Removed + 0.5))) |>
  gf_point(emp_logit ~ Distance, color = ~Morph, ylab = "Empirical logit") |>
  gf_lm()
```



(a) State the hypotheses for the goodness-of-fit test, and give your conclusion in context.

(b) Every row has at least 9 moths removed and 30 not removed. Why does this matter for the test in (a)?

(c) Based on the plot, what is wrong with this model? What would you change?

4. Perform hypothesis tests about a single regression coefficient

Example question Use the model below for whether an article is fake, using `negative` and `title_has_excl`. Before looking at the data, a researcher hypothesized that articles with more negative sentiment are *more* likely to be fake, after accounting for whether the title has an exclamation point.

```
news_add <- glm(type_num ~ negative + title_has_excl, data = news, family = "binomial")
tidy(news_add)
```

```
# A tibble: 3 x 5
  term               estimate std.error statistic  p.value
  <chr>                 <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)          -1.88      0.502     -3.75 0.000180
2 negative              0.383     0.146      2.61 0.00893
3 title_has_exclTRUE    2.53      0.786      3.21 0.00131
```

(a) State the hypotheses for the researcher's question in words and in notation.

(b) Find the p-value for this test, and give your conclusion in context.

(c) What would the p-value be if the researcher had instead hypothesized that articles with more negative sentiment are *less* likely to be fake?

Example question Use the `bud_int` output from Skill 1. Is there evidence that the association between dose and the chance of dying is different for male and female moths?

```
2 * (1 - pnorm(1.31))
```

```
[1] 0.1901958
```

(a) State the hypotheses in words and in notation.

(b) Show where the test statistic in the output comes from, and explain how the p-value is calculated.

(c) Give your conclusion in context.

5. Construct and interpret confidence intervals for regression coefficients and linear combinations of regression coefficients

Example question Use the `med_glm` output from Skill 2, and the `qnorm()` output below.

```
qnorm(0.95)
```

```
[1] 1.644854
```

(a) Calculate a 90% confidence interval for the coefficient of `GPA`.

(b) Interpret your interval on the odds scale, for a 0.1-point increase in GPA.

Example question Use the `bud_int` model from Skill 1. Below is the covariance matrix of the estimated coefficients.

```
round(vcov(bud_int), 4)
```

```
                (Intercept)    sexM log2(dose) sexM:log2(dose)
(Intercept)          0.3055 -0.3055    -0.0839          0.0839
sexM                -0.3055  0.6058     0.0839         -0.1866
log2(dose)          -0.0839  0.0839     0.0279         -0.0279
sexM:log2(dose)      0.0839 -0.1866    -0.0279          0.0729
```

(a) Write the slope of $\log_2$(dose) for **male** moths as a linear combination of the β's, and calculate its estimate.

(b) Calculate the standard error of your estimate in (a).

(c) Calculate a 95% confidence interval for the slope for male moths, and interpret it on the odds scale.

6. Use a regression model to make appropriate predictions

Example question Below is the model from the homework for whether a fatal accident was tire-related (`Cause`), using the age of the vehicle (in years), the number of passengers, and the make of the vehicle (Ford or Other).

```
tire_glm <- glm(Cause ~ VehicleAge + I(VehicleAge^2) + Passengers * Make,
                data = tires, family = "binomial")
tidy(tire_glm)
```

```
# A tibble: 6 x 5
  term                 estimate std.error statistic      p.value
  <chr>                   <dbl>     <dbl>     <dbl>        <dbl>
1 (Intercept)           -10.8       2.23      -4.87 0.00000110
2 VehicleAge              3.42      1.30       2.62 0.00871
3 I(VehicleAge^2)        -0.396     0.194     -2.04 0.0412
4 Passengers              0.839     0.149      5.65 0.0000000159
5 MakeOther              -1.53      0.717     -2.14 0.0326
6 Passengers:MakeOther   -0.636     0.330     -1.93 0.0539
```

(a) Calculate the predicted log-odds, odds, and probability that an accident in a 3-year-old Ford with 2 passengers was tire-related.

(b) Repeat (a) for a 3-year-old vehicle of another make with 2 passengers.

(c) For a Ford with 2 passengers, at what vehicle age are the predicted odds of a tire-related accident the highest?

Example question Below is the model from the homework for whether a family purchases a new car in the next year, using `income` (in thousands of dollars) and `car_age` (the age of their oldest car, in years).

```
cars_glm <- glm(y ~ income + car_age, data = car_purchase, family = "binomial")
tidy(cars_glm)
```

```
# A tibble: 3 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)  -4.74      2.10       -2.25  0.0242
2 income        0.0251    0.0104      2.41  0.0158
3 car_age       0.599     0.390       1.53  0.125
```

(a) Calculate the predicted probability that a family with an income of $150 thousand and an oldest car that is 5 years old will purchase a new car.

(b) For a family whose oldest car is 2 years old, what income gives a predicted probability of 0.5 of purchasing a new car? (*Hint:* what are the odds and log-odds when the probability is 0.5?)

(c) Two families have the same income, but one family's oldest car is 5 years old and the other's is 2 years old. How many times larger are the predicted odds of purchasing a new car for the first family?

7. Conduct inference to compare nested regression models

Example question For the smartphone and sleep data, we want to know whether smoking and drinking help explain sleep quality, after accounting for smartphone dependence, age, BMI, and smartphone use before bed. Below are the residual deviances of the two models and some χ^2 critical values.

```
sleep_reduced <- glm(good_sleep ~ phone_dependence + age + bmi + phone_before_bed,
                     data = sleep, family = "binomial")
sleep_full <- glm(good_sleep ~ phone_dependence + age + bmi + phone_before_bed + smoking + dr
                  data = sleep, family = "binomial")

deviance(sleep_reduced)
```

```
[1] 546.6407
```

```
deviance(sleep_full)
```

```
[1] 542.1046
```

```
qchisq(0.95, df = 1:3)
```

```
[1] 3.841459 5.991465 7.814728
```

(a) Explain why these two models are nested.

(b) State the null and alternative hypotheses.

(c) Calculate the likelihood ratio test statistic and give its degrees of freedom. What do you conclude at $\alpha = 0.05$?

Example question Below is a model for medical school acceptance that adds the applicant's `Sex` to `med_glm` from Skill 2, along with a likelihood ratio test comparing the two models.

```
med_sex <- glm(Acceptance ~ GPA + MCAT + Sex, data = medschool, family = "binomial")
tidy(med_sex)
```

```
# A tibble: 4 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  -24.0       6.97      -3.44 0.000578
```

```
2 GPA            5.14      1.85       2.78 0.00549
3 MCAT           0.181     0.108      1.67 0.0939
4 SexM          -1.26      0.730     -1.72 0.0850
```

```
anova(med_glm, med_sex, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: Acceptance ~ GPA + MCAT
Model 2: Acceptance ~ GPA + MCAT + Sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        52     54.014
2        51     50.786  1   3.2278   0.0724 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) State the hypotheses for the likelihood ratio test, and give your conclusion at $\alpha = 0.05$.

(b) The `SexM` row of the coefficient table also tests whether `Sex` is needed. Compare its p-value to the LRT p-value. Why aren't they exactly the same?

(c) Calculate AIC for both models ($AIC = \text{deviance} + 2k$, where k is the number of coefficients). Which model does AIC prefer? Does this contradict your answer to (a)?

8. Identify a binomial logistic regression scenario and whether overdispersion is present

Example question For each scenario, say whether you would use binary logistic regression, binomial logistic regression, or neither. For the binomial scenarios, what is n_i?

(a) Each row is a news article, and the response is whether it is fake or real.

(b) Each row is a batch of 20 budworm moths, and the response is the number in the batch that died.

(c) Each row is a child in the language exposure study, and the response is the number of the 4 trials the child got right.

(d) Each row is a forest site, and the response is the number of possum species found there.

(e) Each row is a student, and the response is whether they have good sleep quality (a PSQI of 5 or below).

(f) Each row is a moth in the budworm experiment, and the response is whether it died. Which model is this, and how is it related to (b)?

Example question Below are the Pearson χ^2 statistics and residual degrees of freedom for two binomial logistic regression models: the budworm model `sex + log2(dose)`, and the language exposure model from Skill 2.

```
bud_add <- glm(cbind(numdead, N - numdead) ~ sex + log2(dose),
               data = budworm, family = "binomial")
sum(residuals(bud_add, type = "pearson")^2)
```

```
[1] 5.306017
```

```
df.residual(bud_add)
```

```
[1] 9
```

```
sum(residuals(lang_glm, type = "pearson")^2)
```

```
[1] 143.4482
```

```
df.residual(lang_glm)
```

```
[1] 61
```

(a) Estimate the dispersion parameter $\widehat{\psi}$ for each model. Which model shows evidence of overdispersion?

(b) For the language model, the binomial z-test for `ppvt` gives $z = -1.71$ (p $= 0.087$). Calculate the test statistic for `ppvt` under the quasibinomial model. Would your conclusion at $\alpha = 0.10$ change?

(c) Give one reason why the language data might be overdispersed when the budworm data are not.

9. Poisson regression

Example question The `equality_index` data records the number of anti-discrimination `laws` in each US state. The model below uses the percent of the state's population that lives in urban areas (`percent_urban`) and the state's `region` (`midwest` is the reference level).

```r
eq_glm <- glm(laws ~ percent_urban + region, data = equality_index, family = "poisson")
tidy(eq_glm)
```

```
# A tibble: 5 x 5
  term            estimate std.error statistic  p.value
  <chr>              <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)      -1.51     0.302      -4.99  6.03e- 7
2 percent_urban     0.0458   0.00376    12.2   4.69e-34
3 regionnortheast   0.643    0.148       4.33  1.47e- 5
4 regionsouth       0.0964   0.151       0.639 5.23e- 1
5 regionwest        0.918    0.136       6.74  1.58e-11
```

(a) Using proper notation, write the theoretical Poisson regression model that was fit. Define all variables.

(b) Interpret the coefficient for `percent_urban` for a 10-percentage-point increase, and the coefficient for `regionwest`, in context.

(c) Calculate the predicted mean number of laws for a midwest state where 70% of the population lives in urban areas.

(d) Below is a likelihood ratio test for whether `region` is needed. State the hypotheses, the test statistic and degrees of freedom, and your conclusion.

```r
eq_reduced <- glm(laws ~ percent_urban, data = equality_index, family = "poisson")
anova(eq_reduced, eq_glm, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: laws ~ percent_urban
Model 2: laws ~ percent_urban + region
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        48     601.82
2        45     525.76  3   76.058 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(e) The sum of the squared Pearson residuals for `eq_glm` is 723.53 on 45 df. What does this tell you? How would the standard errors change if we used a quasipoisson model?

Example question Below is a Poisson regression model for the number of possum species found at a forest site (`y`), using the basal area of acacia (`Acacia`), the number of shrubs (`Shrubs`), and whether stumps from logging are present (`Stumps`, 1 = yes).

```r
possum_glm <- glm(y ~ Acacia + Shrubs + Stumps, data = possums, family = "poisson")
tidy(possum_glm)
```

```
# A tibble: 4 x 5
  term        estimate std.error statistic   p.value
  <chr>          <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)  -0.300    0.206       -1.46 0.145
2 Acacia        0.0323   0.00815      3.96 0.0000749
3 Shrubs        0.0437   0.0187       2.34 0.0194
4 Stumps       -0.398    0.272       -1.46 0.144
```

(a) Interpret the coefficient for `Stumps` in context.

(b) Calculate a 95% confidence interval for the coefficient of `Acacia`, and interpret it for a 10-unit increase in `Acacia`.

(c) The sum of the squared Pearson residuals is 141.02 on 147 df. Is there evidence of overdispersion? Should we use the Poisson or the quasipoisson model?

(d) One of the Poisson model's assumptions is that the mean equals the variance. Explain what this means in the context of these data.